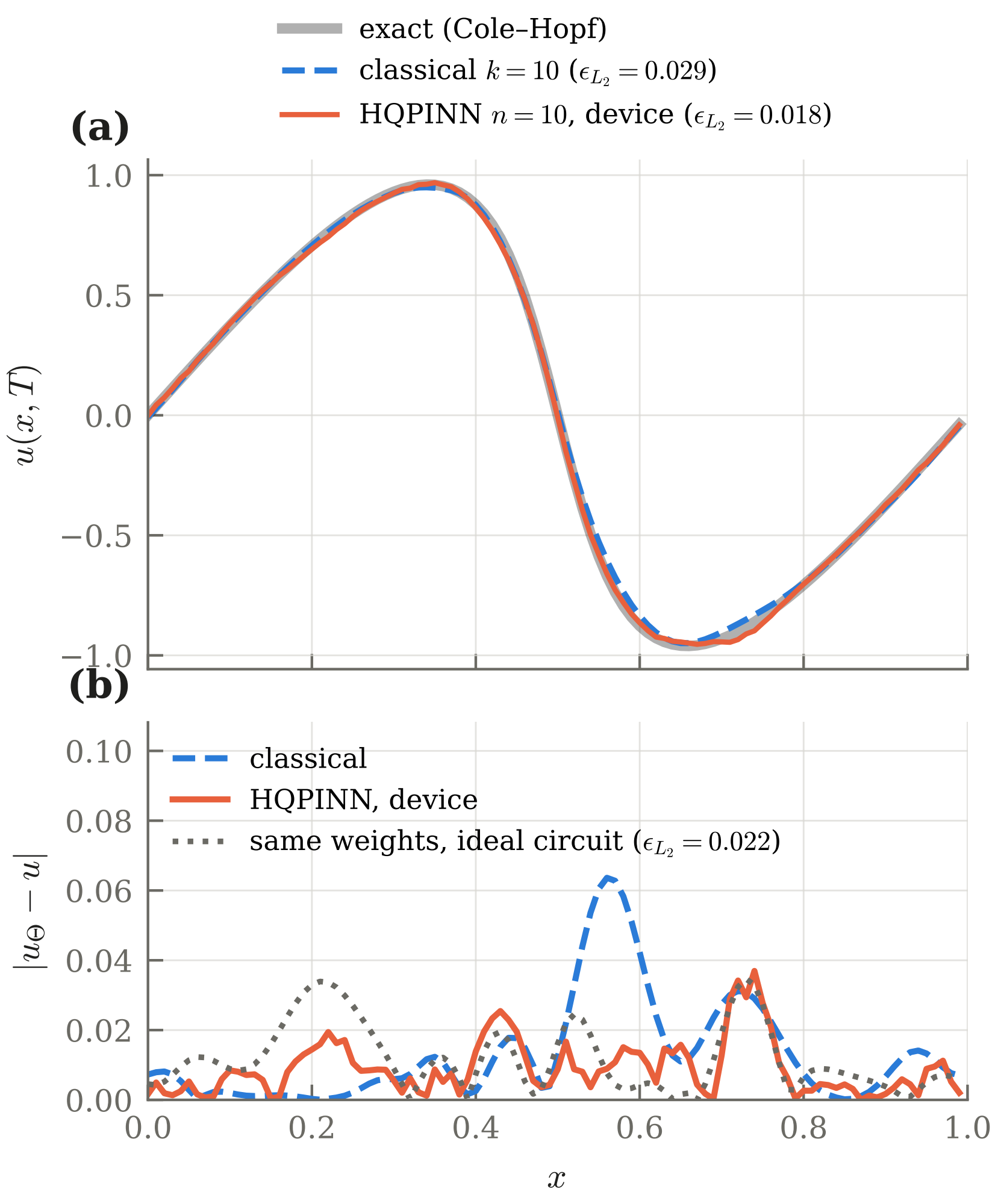
exact (Cole–Hopf)
classical $k = 10$ ($\epsilon_{L_2} = 0.029$)
HQPINN $n = 10$, device ($\epsilon_{L_2} = 0.018$)
(a)
1.0
0.5
0.0
−0.5
−1.0
$u(x, T)$
(b)
classical
HQPINN, device
same weights, ideal circuit ($\epsilon_{L_2} = 0.022$)
0.10
0.08
0.06
0.04
0.02
0.00
$|u_\Theta - u|$
0.0
0.2
0.4
0.6
0.8
1.0
x